

Yashi Srivastava

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🌐 LinkedIn — 🐙 GitHub — </> LeetCode

Education

LNMIIT Institute of Information Technology (LNMIIT), Jaipur

2023 – 2027

B.Tech, Communication & Computer Engineering — Specialisation: AI & Data Science

Experience

Research Intern — MNNIT Allahabad

June 2025 – July 2025

Deep Learning — Supervisor: Dr. Joohi Chauhan

[Certificate]

- Trained ANN and CNN models in **PyTorch** on structured (NHANES) and image (CIFAR-10) datasets; achieved **86.5% accuracy & 0.89 ROC-AUC** on cardiac risk prediction and **74.2% accuracy** on CIFAR-10 classification.
- Applied regularization (dropout, batch norm), class-weighted loss for imbalanced data, and benchmarked Adam/RMSProp/SGD optimisers using ROC-AUC, precision-recall, and F1-score.
- Studied **federated learning** (FedAvg) for privacy-preserving distributed ML; reviewed gradient aggregation and communication efficiency tradeoffs.

Projects

🔗 **Barbell Exercise Classification & Rep Counter**

Python, Scikit-learn, PCA, Pandas

- Processed 6-axis IMU sensor time-series data; engineered temporal/frequency features and applied **PCA** (95% variance retained) to reduce dimensionality before classification.
- Compared Random Forest, SVM, and KNN classifiers with **GridSearchCV** (5-fold CV); built automated rep counter via peak detection, improving accuracy ~**12% over baseline**.

🔗 **AI-Assisted Symbolic Execution & Bug Detection Framework**

C/C++, KLEE, Z3, LLVM, Python

- Built a symbolic execution pipeline using **KLEE** (LLVM-based) and **Z3 SMT Solver** to explore all program paths and auto-detect bugs (null dereferences, buffer overflows, divide-by-zero).
- Integrated AI-based patch suggestion module; achieved significantly higher branch coverage than random test generation on benchmark C programs.

🔗 **Cardiac Risk Prediction using ANN**

PyTorch, Scikit-learn, Pandas, Matplotlib

- Built multi-layer ANN on **NHANES** dataset for 10-year CVD risk prediction; handled class imbalance via weighted cross-entropy. **86.5% accuracy — 85.0% precision — 72.3% recall — 0.89 AUC**.

🔗 **CIFAR-10 Image Classification using CNN**

PyTorch, NumPy, Matplotlib

- Designed CNN with Conv-Pool blocks, dropout, and batch norm; applied data augmentation and cosine LR scheduling. **Achieved 74.2% test accuracy** on 60,000-image dataset over 50 epochs.

Technical Skills

- **Languages:** Python, Java, C, JavaScript, HTML/CSS, SQL
- **ML/DL:** PyTorch, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- **Concepts:** Supervised Learning, CNNs, ANNs, Federated Learning, Feature Engineering, Hyperparameter Tuning, Class Imbalance Handling
- **Tools:** Git/GitHub, KLEE, Z3, LLVM, LaTeX

Certifications & Achievements

- **Selected Participant, ACM-W India Grad Cohort 2025, IIT Kanpur** — competitive program for undergraduate women in computing.
- **INSPIRE Scholarship, Govt. of India** — awarded to top 1% of national 12th Board scorers for excellence in S&T.
- **Qualified Stage 1, NTSE (National Talent Search Examination)**.

Relevant Coursework

- DSA • OOP • DBMS • Operating Systems • Computer Networks • AI & Machine Learning • Software Engineering

Positions of Responsibility

- **Chairperson, ACM Student Chapter — LNMIIT**